Name
Date
Course

Sequences and Series (Homework)

Find the next three terms in each sequence.

1. $\frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \ldots$
2. $100\ 36, -28, -92, \ldots$
3. $800, 100, \frac{25}{2}, \frac{25}{16}, \ldots$
4. $-20, 40, -80, 160, \ldots$

Find the indicated term in each arithmetic sequence below.

5. $2, 2.9, 3.8, \ldots$
   $n = 10$
6. $-\frac{1}{3}, -\frac{1}{6}, 0, \ldots$
   $n = 21$
7. $-3, -9, -15, \ldots$
   $n = 42$
8. $632, 31, -570, \ldots$
   $n = 89$

Find the sum of the first "n" terms of each arithmetic sequence.

9. $-19, -11, -3, \ldots$
   $n = 27$
10. $0.007, 0.003, -0.001, \ldots$
    $n = 80$

11. $20, 17, 14, \ldots$

$n = 19$

Find the indicated term in each geometric sequence below. You may leave your answers in exponential form if they are very large or very small.

12. $-\frac{1}{20}, -\frac{1}{4}, -\frac{5}{4}, \ldots$

$n = 7$

13. $48, 12, 3, \ldots$

$n = 24$

14. $-27, 36, -48, \ldots$

$n = 13$

15. $1, 0.4, 0.16, \ldots$

$n = 49$

Find the sum of the first "n" terms in each geometric sequence. You may leave your answers in exponential form if they are very large or very small.

16. $\frac{5}{6}, \frac{5}{12}, \frac{5}{24}, \ldots$

$n = 14$

17. $2, -1.8, 1.62, \ldots$

$n = 25$

18 $1, 3, 9, \ldots$

$n = 10$

19 $-0.3, -0.6, -1.2, \ldots$

$n = 8$

Answers

1. $\boxed{\frac{1}{4}, \frac{1}{3}, \frac{1}{2}}$
2. $\boxed{-156, -220, -284}$
3. $\boxed{\frac{25}{128}, \frac{25}{1024}, \frac{25}{8192}}$
4. $\boxed{-320, 640, -1280}$
5. $\boxed{10.1}$
6. $\boxed{3}$
7. $\boxed{-249}$
8. $\boxed{-52,256}$
9. $\boxed{2,295}$
10. $\boxed{-12.08}$
11. $\boxed{-133}$
12. $\boxed{-781.25}$
13. $\boxed{48\left(\frac{1}{4}\right)^{23}}$ or $\boxed{3\left(\frac{1}{4}\right)^{21}}$
14. $\boxed{-27\left(-\frac{4}{3}\right)^{12}}$ or $\boxed{-\frac{4^{12}}{3^9}}$
15. $\boxed{(0.4)^{48}}$
16. $\left(\frac{5}{6}\right)\left(\frac{1-\left(\frac{1}{2}\right)^{14}}{1-\frac{1}{2}}\right) = \boxed{\left(\frac{5}{3}\right)\left(1-\left(\frac{1}{2}\right)^{14}\right)}$
17. $(2)\frac{1-(-0.9)^{25}}{1-(-0.9)} = \boxed{2\,\frac{1-(-0.9)^{25}}{1.9}}$
18. $\frac{1-3^{10}}{1-3} = -\frac{1-3^{10}}{2}$ or $\boxed{29,524}$
19. $-0.3\frac{1-2^8}{1-2} = \boxed{-76.5}$